

UĞUR ARIKAN

RUST SOFTWARE ENGINEER | TECHNICAL LEAD

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Software engineer and technical lead with a focus on Rust, high-performance systems, and reusable software for computationally demanding applications. I design concurrent and parallel execution components and build APIs and visualization layers used in decision-support workflows. My background in operations research and data science adds depth in algorithms, mathematical modeling, and complex systems.

CORE AREAS

Rust engineering
Systems and libraries
Technical leadership

Parallel computation
Concurrency and performance
Production APIs

Optimization algorithms
Applied data science
Open-source contribution

PROFESSIONAL EXPERIENCE

Principal Operations Research & Data Scientist | DHL Data & AI | Bonn, Germany **2018 - Present**

- Design and deliver production software for computationally demanding optimization and decision-support applications, from algorithm implementation through deployment and operational use.
- Develop decision-support tools and optimization workflows for large-scale transportation and logistics networks, supporting capacity, routing, and planning decisions in complex operational settings.
- Build scalable services and reusable components in Rust, Python, C#, and TypeScript, covering algorithm engineering, asynchronous APIs, visualization, and deployment.
- Work closely with business, network, and technical stakeholders to translate planning questions into tractable mathematical models and usable decision-support applications.
- Lead the adoption of Rust for optimization-focused software, establishing production-oriented practices for performance, testing, maintainability, and reuse.
- Mentor colleagues and collaborate with technical, analytical, and operational stakeholders to turn complex requirements into maintainable software systems.

Postdoctoral Researcher | Singapore University of Technology and Design | Singapore **2014 - 2018**

- Conducted research in transportation systems, utility theory, discrete choice models, and robust optimization; supervised student projects and contributed to multidisciplinary design teams.

SOFTWARE ENGINEERING & OPEN SOURCE

- Apply parallel computation, concurrency, profiling, and benchmarking to high-performance numerical and optimization workloads, designing reusable components and scalable software architectures.
- Publish and maintain a collection of public orx-* Rust crates, including the orx-parallel library for parallel computation, as well as crates for concurrent data structures, general-purpose data structures, closure utilities, network-flow optimization, and other numerical methods. The projects emphasize reusable abstractions, documentation, testing, and benchmarking, and are published via crates.io.

EDUCATION

PhD, Industrial Engineering & Operations Research | Middle East Technical University **2009 - 2014**

Dissertation: Airline Disruption Management

MSc, Industrial Engineering & Operations Research | Middle East Technical University **2006 - 2009**

Dissertation: Two-Sided Assembly Line Balancing: Models and Heuristics

BSc, Industrial Engineering & Operations Research | Middle East Technical University, **2002 - 2006**

Ankara, Turkey

FURTHER TRAINING

Modeling and Simulation of Transportation Networks | Massachusetts Institute of **Jul 2015**

Technology

Operations Research Peripatetic Postgraduate Program | JKU Linz **Jul 2013**

RESEARCH & PUBLICATIONS

Research and peer-reviewed publications in airline disruption management, network-based recovery, traffic equilibrium, and applied optimization in logistics and transportation.

- **Flight Network-Based Approach for Integrated Airline Recovery with Cruise Speed Control**
INFORMS Transportation Science
- **Integrated Aircraft and Passenger Recovery with Cruise Time Controllability**
Annals of Operations Research
- **On the flexibility of using marginal distribution choice models in traffic equilibrium**
Transportation Research Part B: Methodological
- **Distributionally Robust Markovian Traffic Equilibrium**
INFORMS Transportation Science
- **Human-Centric Parcel Delivery with Operations Research and Machine Learning**
INFORMS Interfaces

SELECTED TALKS & TECHNICAL WRITING

- **Composing Zero-Cost Abstractions in Route Optimization**

Talk on the intersection of Rust and optimization

- **orxfun notes**

Technical writing on Rust and software engineering, including publications in This Week in Rust

SELECTED TECHNICAL CAPABILITIES

OPTIMIZATION

Mixed-integer optimization
Network optimization
Routing and facility location
Heuristics, decomposition, and local search
Scenario analysis and decision support

ENGINEERING

Rust, Python, C#, TypeScript/React, SQL
Parallel computation and concurrency
Production service design
Cloud deployment and integration
Profiling, benchmarking, and reusable library design

LANGUAGES

English (fluent) | German (intermediate) | Turkish (native)